

Relationship between difference of preoperative and cardiopulmonary bypass mean arterial pressure, and acute kidney injury in cardiac surgical patients undergoing valve surgery

Anity Singh Dhanyee,¹ Satyen Parida,² 
Chitra Rajeswari Thangaswamy,² Ajay Kumar Jha,²
Medha Rajappa,³ Hemachandren Munuswamy⁴ and
Sandeep Kumar Mishra²

Abstract

Background: Modifiable and non-modifiable factors contribute to development and progression of acute kidney injury (AKI) during cardiac surgery. We hypothesized that, the difference between preoperative mean arterial pressure (MAP) and the average mean arterial pressure maintained on cardiopulmonary bypass (CPB) would be strongly predictive of AKI. We also measured plasma Neutrophil gelatinase-associated lipocalin (NGAL), to establish its association with cardiac surgery associated-AKI (CSA-AKI).

Methods: One hundred and twelve high-risk patients undergoing valve, and valve plus coronary artery bypass grafting (CABG) surgery under cardiopulmonary bypass (CPB) were included in this study. Delta mean arterial pressure (MAP) was calculated as the difference between the average of pre-operative and on-bypass MAP, and blood was sampled for NGAL levels, at baseline, and 6-h after CPB. Detailed data collection was done, tabulating most of the factors which might influence development of post-operative cardiac surgery associated-AKI (CSA-AKI). To define CSA-AKI within the first 24-h post-operatively, the Kidney Disease Improving Global Outcomes (KDIGO) classification was used.

Results: Out of 112 patients, 44 (39.3%) developed CSA-AKI postoperatively. With an ROC analysis cut-off of delta MAP of more than 25.67 mmHg, 46.4% patients developed post-operative AKI, and the average CPB flows which were 1.8 ± 0.2 were not contributory to the development of early CSA-AKI. In our study, ELISA test for human NGAL was performed on serum samples, and the estimated cut-off value of 1661 ng/mL was found to be significantly associated with early CSA-AKI.

Conclusions: Delta MAP and CPB flows are not related to early post-surgical CSA-AKI in cases with prior high-risk elements. However, baseline serum NGAL, as well as its percent change during the early post-surgical period independently predicted the development of CSA-AKI. This implies that, there may be patients with a higher pre-operative preponderance to develop this complication, which could actually be delineated by the use of serum NGAL estimations at baseline.

Keywords

delta mean arterial pressure, cardiopulmonary bypass flow, neutrophil gelatinase-associated lipocalin, cardiac surgery associated-acute kidney injury post-operatively

Introduction

Acute kidney injury (AKI) complicates the postoperative course after cardiac surgery in up to 30% of patients, with 1%–2% requiring renal replacement therapy, which incurs increased mortality and morbidity.¹ Various factors, both modifiable and non-modifiable, contribute to the development and progression of AKI. It is seen that many of those who

¹Department of Anaesthesiology & Critical Care, Sri Balaji Vidyapeeth (Deemed-to-be-University), Mahatma Gandhi Medical College & Research Institute, Puducherry, India

²Department of Anaesthesiology & Critical Care, JIPMER, Puducherry, India

³Department of Biochemistry, JIPMER, Puducherry, India

⁴Department of Cardiothoracic and Vascular Surgery, JIPMER, Puducherry, India

Corresponding author:

Satyen Parida, Dept of Anaesthesiology & Critical Care, JIPMER, Qr. No. Type VB-2, JIPMER Campus, Dhanvantari Nagar, Puducherry 605006, India.

Email: jipmersatyen@gmail.com

recover from renal replacement therapy or even mild AKI, progress to chronic kidney disease in the ensuing months and years. Hence, it is possible that the difference between mean arterial pressure obtained preoperatively, and that maintained during CPB, might be closely related to the incidence of CSA-AKI. Also time-sensitive biomarkers like serum NGAL might help predict AKI prior to its actual occurrence in these patients. A study done earlier has related the delta MAP (decrease in on-bypass MAP from baseline) to an increase in incidence of AKI in a select population of patients who were at high-risk for AKI.² Therefore, we conducted a prospective observational trial using a cohort of patient's undergoing valvular heart surgery, with risk factors for renal dysfunction. Our primary objective was to study the relationship between delta MAP and the incidence of cardiac surgery-associated-AKI (CSA-AKI) 24-h post-operatively using the Kidney Disease Improving Global Outcomes (KDIGO) classification. The secondary objectives included evaluating the effects of cardiopulmonary bypass (CPB) flows, level of biomarker plasma Neutrophil gelatinase-associated lipocalin (NGAL) and various other post-operative variables on the incidence of CSA-AKI 24-h postoperatively.

Methods

Study design and population

This study was designed as a prospective observational cohort study. It was conducted in the department of Anesthesiology and Critical Care, JIPMER, Pondicherry, India, after approval from the Institute Ethics Committee (Human studies) - JIP/IEC/2020/075. The trial was also registered in the Clinical Trials Registry of India - CTRI/2020/08/027448. Consecutive patients, more than 18 years of age scheduled for elective valve, and combined valve and coronary artery bypass grafting (CABG) surgery through median sternotomy on cardiopulmonary bypass were screened from Aug 2021 to June 2022. Written informed consent explaining the procedure, risks, and adverse events was obtained before enrollment into the study. The inclusion criteria for the study included patients above 18 years of age, scheduled for elective valve surgery or CABG with valve surgery on CPB, who had at least one additional feature, putting them at risk for CSA-AKI. These additional risk factors included: age ≥ 70 years, diabetes mellitus (DM), history of congestive heart failure or documented LVEF $< 35\%$, New York Heart Association (NYHA) symptom severity class 3 or 4, pre-operative serum creatinine ≥ 1.2 mg/dl; and

previous cardiac surgery. Exclusion criteria were end-stage kidney disease (CKD class V) and prior kidney transplantation.

Study definitions and outcomes

Primary outcome definition

Acute kidney injury (kidney disease improving global outcomes classification)

Stage 1- Serum creatinine 1.5–1.9 times baseline or >0.3 mg/dl (>26.5 $\mu\text{mol/l}$) increase and urine output <0.5 mL/kg/h for 6–12 h;

Stage 2- Serum creatinine 2.0–2.9 times baseline and urine output <0.5 mL/kg/h for >12 h; and

Stage 3- Serum creatinine 3.0 times baseline or increase in serum creatinine to >4.0 mg/dl (>353.6 $\mu\text{mol/l}$) or initiation of renal replacement therapy or in patients <18 years, decrease in estimated glomerular filtration rate (eGFR) to <35 mL/min per 1.73 m² and urine output <0.3 mL/kg/h for >24 h or anuria for >12 h.

Secondary outcomes definition. Mortality - Death within 30 days of surgery; intra-aortic balloon pump (IABP) placement for low cardiac output state (LCOS) - transient decrease in systemic perfusion secondary to myocardial dysfunction; significant inotrope requirement - requirement of inotropes even after 24 h of completion of surgery; and post-operative renal replacement therapy (RRT) - up to 5 days post-operatively; and plasma NGAL levels - at baseline and 6 h after commencement of CPB.

We followed up patients for development of CSA-AKI within the first 24-h post-operatively after cardiac surgery, to capture the AKI most likely attributable to CPB exposure and intra-operative factors, rather than factors in the postoperative period. For measuring plasma NGAL, we obtained blood samples at baseline and 6 h after commencement of CPB. We centrifuged these samples at 2000 rpm for 5 min and stored the supernatant in equal volumes at -80°C until measurement. Serum creatinine was measured pre-operatively and on the first day post-operatively utilizing the Beckmann coulter model Au5800 by modified Jaffe's method standardized by isotope dilution mass spectroscopy. The inter- and intra-assay coefficients of variation of plasma NGAL and creatinine measurement were $<5\%$.

Study protocol

Detailed manual data collection was done using standardized case-report forms from the trends of the

multiparameter monitor, at every 15-min intervals and entered into an Excel sheet for analysis. The following data was tabulated: demographic profile including patient's age, gender, co-morbidities, medications, baseline kidney function; surgical details like type of surgery, approach, aortic cross-clamp time, total bypass time, intraoperative parameters like on-bypass mean arterial pressure, pump flow rate, ultrafiltration, temperature, hemoglobin, blood transfusions, use of vasopressors, anti-fibrinolytic dose, post-operative laboratory investigations and ELISA test for baseline and 6-h serum NGAL post-CPB. Delta MAP was calculated as the average MAP pre-operatively (2 readings using automated blood-pressure cuff at pre-anesthetic check-up and on the day of surgery and another reading from the radial arterial line before induction of anesthesia) minus average MAP on-bypass (readings at every 15-min interval during CPB). Clinical outcomes data was also collected including development of AKI post-operatively, requirement of continuous renal replacement therapy (CRRT), days on mechanical ventilation, length of hospital stay and mortality.

Surgical procedure and cardiopulmonary bypass

The following steps enumerate routine clinical care, which is followed for all cardiac surgeries in our institution. All study patients underwent surgery via midline sternotomy and on cardiopulmonary bypass. Preoperative MAP was documented for analysis every 5 min prior to going on CPB. CPB was instituted using standard techniques with cannulation of the ascending aorta with an aortic cannula and a bi-caval or two-stage single venous cannulation technique employed for venous drainage, as the case demanded. A membrane oxygenator and roller pump were used in all patients. The circuit was primed with crystalloids, mannitol up to 1400–1800 mL, typically representing 30%–35% of the patient's blood volume, and unfractionated heparin (UFH) 5000 IU as per protocol. A nasopharyngeal probe was placed to monitor temperature throughout surgery and on CPB, moderate hypothermia, with temperature around 28°C was maintained. Prior to cannulation of the ascending aorta, unfractionated heparin 400 units/kg was given intravenously. Activated clotting time was maintained more than 480 s throughout bypass. The bypass pump flow was maintained around 2.4 L/min/m² and the perfusion pressure 50–60 mmHg, was modified as per the requirement of surgery, to provide a bloodless surgical field and to maintain adequate organ perfusion. If required, vasopressor in the form of phenylephrine infusion, 20 µg/mL was used to

maintain the perfusion pressure at greater than 50 mmHg. The bypass circuit was primed and filled with Ringer Lactate solution. Blood was transfused on CPB whenever hemoglobin dropped below 7 g/dL. To achieve myocardial protection, blood cardioplegia in St Thomas solution (ratio of 4:1) with high potassium (20 mmol/L) at induction, and low potassium (8 mmol/L) for maintenance (blood-crystalloid ratio 16:1) was used. Cardioplegia delivery was via one of the three approaches: antegrade via aortic root, direct cannulation of the coronary ostia or retrograde via the coronary sinus. After completion of the surgical procedure and closure of the heart in valve replacement procedures, the heart and lungs were passively filled with blood from the CPB circuit. The heart was massaged gently, and the left side was de-aired continuously through an appropriate vent. The aortic cross clamp was then released, and the vent was opened again. After the procedure, following decannulation, heparin was reversed with protamine at a dose 1.3 times the original dose of heparin administered. After surgery, patients were transferred to the cardiothoracic surgical intensive care unit for postoperative management and recovery. As per the weaning protocol followed in the intensive care unit (ICU), patients were extubated at the discretion of the ICU consultant.

Statistical analysis

The study was planned with an α of 95%, power of 80%, unexposed to exposed ratio of 1, and expected incidence in unexposed and exposed at 20% and 41% respectively, considering exposure to be a decline in delta MAP of 26%, based on a previous trial,² which gave a sample of 112 by Kelsey method (calculated using <https://www.openepi.com/Samplesize/SSCohort.htm>). The sampling technique followed was convenience sampling.

SPSS (Statistical Package for the Social Studies) Statistics for Windows, version 21.0 (IBM Corp., Armonk, N.Y., USA) was used to perform the statistical analysis. The primary outcome was the incidence of CSA-AKI, defined by KDIGO classification. Patient demographic, clinical, physiologic and laboratory data for the pre- and intra-operative periods were summarized as mean ($\pm$ SD) or median (interquartile ranges [IQR]), and numbers with percentage. Comparison was done using Wilcoxon rank tests, t-tests and chi-square tests, as deemed appropriate. Any missing data was not replaced. We evaluated delta MAP and CPB flow both as continuous variables and then dichotomized them using an outcome-based cut-off method. Correlation analysis was used for analyzing the relationship between plasma NGAL and delta MAP. A p -value < .05 was considered statistically significant. Furthermore, data was presented

as odds ratios (OR) with 95% confidence intervals (CI). Those factors found significant by univariate analysis ($p < .2$) were candidates for multivariable logistic regression.

Results

Patient characteristics

Of the 161 patients over the age of 18 years screened from August 2021 to June 2022, 112 met our inclusion criteria and were analyzed. We excluded 49 patients from this study because they either did not undergo valve or valve plus CABG procedure, or could not have their NGAL levels assessed owing to logistic reasons. A total of 44 patients developed CSA-AKI, an incidence of 39.3%. Out of 112 recruited patients, 62% were male and 38% female, with 95% being American Society of Anesthesiologists (ASA) III and 5% ASA IV. Those patients developing CSA-AKI were more likely to be older, with higher body mass index (BMI), have diabetes mellitus (65% vs 35%, $p = .009$) and/or previous myocardial infarction (71.4% vs 28.6%, $p = .008$) as compared to the non-AKI group. There was a significant difference between patients taking or not taking clopidogrel (66.7% vs 33.3%, p value $< .04$) with regard to post-operative AKI. This probably was secondary to the above-mentioned co-morbidities. The fact that our study cohort belongs to a high-risk group for postoperative CSA-AKI, was also re-confirmed through this analysis (Table 1).

Association of Acute kidney injury with intra-operative parameters

The placement of the side biter clamp (72.7% vs 27.3%, p value $< .023$) and higher number of grafts (3.0 ± 1.1 vs 1.0 ± 1.1 , p value $< .048$) placed in combined valve and CABG surgery showed a significant association with development of postoperative AKI. Higher volume of fluid removed by on-bypass (conventional ultrafiltration = 1424.5 ± 1124.7 vs 984.6 ± 1045.9 , p value $< .037$) and post-bypass (modified ultrafiltration, 164.8 ± 287.4 vs 73.5 ± 160.8 , p value $< .034$) methods of ultrafiltration was also correlated with a higher incidence of CSA-AKI (Table 2).

Association of acute kidney injury with neutrophil gelatinase-associated lipocalin values

Table 3 displays parameters related to serum NGAL values. The serum NGAL-related measurements

including baseline and 6-h post-bypass, both showed significant association with CSA-AKI.

Association of acute kidney injury with postoperative outcomes

Post-operative outcomes and their relation to the development of CSA-AKI are depicted in Table 4. The various post-operative outcomes in the groups with and without AKI, like mortality, use of IABP, post-operative low cardiac output state (LCOS), requirement of CRRT, post-operative central nervous system (CNS) and respiratory system morbidity, wound infection, rate of re-exploration, post-operative arrhythmias, duration of mechanical ventilation and ICU stay did not show any significant association with CSA-AKI.

Receiver-operating characteristic (ROC) curve analysis

Univariate analysis showed that AKI showed significant associations with age, BMI, co-morbidities like diabetes mellitus, previous myocardial infarction (MI), drugs like clopidogrel, and the baseline and post-bypass NGAL, as well as the delta NGAL values (Table 5). The multivariate analysis of the most significant variables, allowing the model to only retain these variables with the greatest association level, shows that only delta NGAL was found to be significantly associated with post-operative CSA-AKI (Table 6).

ROC curves were used to assess the areas under curve (AUC) for delta MAP and delta NGAL. The delta MAP demonstrated an AUC of only 0.569 (Figure 1(a)). However, the AUC on the ROC curve for delta NGAL was 0.942 (Figure 1(b)), demonstrating a high predictive value for AKI. The most accurate cut-off value to distinguish non-AKI and AKI groups was a delta NGAL of 1661.14 ng/mL at sensitivity of 86.4% and specificity of 92.7%. At this threshold, positive likelihood ratio was 11.84 and negative likelihood ratio was 0.15.

Discussion

This prospective observational cohort study was carried out in 112 adult patients, who underwent valve and combined valve and CABG surgery on cardiopulmonary bypass. There are various studies done earlier, which have found that delta MAP and CPB flows maintained during cardiac surgery are independently associated with early postoperative AKI in high-risk patients.^{2,3} These are potentially modifiable intra-operative factors

Table 1. Baseline characteristics of the study population.

	AKI (44)	No AKI (68)	p value
Age in years (mean $\pm$ SD)	45.6 $\pm$ 12.5	38.2 $\pm$ 13.8	.005*
Male gender, n (%)	17 (38.6)	26 (38.2)	.966
ASA status, n (%)			
ASA III	41 (93.2)	65 (95.6)	.678
ASA IV	3 (6.8)	3 (4.4)	
BMI (kg/m ²) (mean $\pm$ SD)	23.5 $\pm$ 3.92	21.6 $\pm$ 4.5	.023*
BSA (m ²) (mean $\pm$ SD)	1.6 $\pm$ 0.2	1.6 $\pm$ 0.2	.059
EUROscore II (mean $\pm$ SD)	1.83 $\pm$ 0.91	1.60 $\pm$ 0.93	.211
Diabetes mellitus, n(%)	13 (29.5)	7 (10.3)	.009*
Hypertension, n (%)	6 (13.6)	8 (11.8)	.770
Previous myocardial infarction, n (%)	10 (22.7)	4 (5.9)	.008*
Dyslipidaemia	3 (6.8)	3 (4.4)	.678
Cerebrovascular disease	2 (4.5)	2 (2.9)	0.645
Peripheral vascular disease	6 (13.6)	4 (5.9)	0.645
Chronic kidney disease	0 (0)	1 (1.5)	1.000
Heart failure	12 (27.3)	28 (41.2)	.134
Rhythm disturbance	18 (40.9)	31 (45.6)	.626
Ejection fraction (%) (Mean $\pm$ SD)	59.2 $\pm$ 7.5	59 $\pm$ 7.5	.860
Baseline serum creatinine in mg/dl (mean $\pm$ SD)	0.76 $\pm$ 0.47	0.70 $\pm$ 0.25	.336
Baseline average MAP in mm Hg (mean $\pm$ SD)	90.3 $\pm$ 9.3	88 $\pm$ 8.7	.180
Aspirin, n (%)	16 (36.4)	19 (27.9)	.348
Clopidogrel, n (%)	8 (18.2)	4 (5.9)	.040*
Beta-blockers, n (%)	35 (79.5)	48 (70.6)	.291
Calcium channel blockers, n (%)	6 (13.6)	7 (10.3)	.590
Diuretics, n (%)	41 (93.2)	67 (98.5)	.136
ACE -inhibitors, n (%)	4 (9.1)	13 (19.1)	.149
Angiotensin receptor blockers, n (%)	3 (6.8)	1 (1.5)	.298
Statins, n (%)	9 (20.4)	8 (11.8)	.211
Heparin, n (%)	42 (95.4)	65 (95.6)	1.000

AKI: Acute Kidney Injury; SD: Standard Deviation; ASA: American Society of Anesthesiologists; BMI: Body Mass Index; BSA: Body Surface Area; MAP: Mean Arterial Pressure; ACE: Angiotensin Converting Enzyme.

* $p \leq .05$.

that might influence the morbidity associated with CSA-AKI that necessitates further investigation. Hence, we wanted to evaluate how significant the correlation between the decrease in MAP on-bypass and the occurrence of postoperative AKI in our high-risk population of patients undergoing valve surgeries might be. The cohort which we have enrolled belongs to a category at high-risk for developing AKI, as was re-confirmed in our analysis. In those who developed AKI post-operatively, we identified a similar gender distribution (p -value 0.966), but the mean age (45.6 ± 12.5 years) in the participants who developed AKI was found to be statistically significantly higher as compared to those in the non-AKI group (38.2 ± 13.8 years; $p = .005$). Our primary objective was to assess the effect of delta MAP on the incidence of AKI 24 h post-operatively, using the KDIGO classification.

In a study done by Kanji et al., a delta MAP of 26 mmHg was independently linked with CSA-AKI in 41% of patients who developed early AKI.² But we found that, out of 112 patients, overall, 44 (39.3%) developed CSA-AKI and the delta MAP was not statistically significantly different between the group that developed AKI as compared to the one that did not have AKI. Similar findings were reported in a few other studies. In a binational retrospective cohort study, Turner et al. concluded that neither the duration spent on low perfusion flow (1.6 L/minute/m^2) nor the length of time at low perfusion pressure ($<50 \text{ mmHg}$) was linked to postoperative AKI.⁴ Other studies by Slogoff et al.⁵ and Sirvinskis et al.⁶ demonstrated similar findings as in our study. Our results are also consistent with recently published ANZCPR findings by Newland et al., who in a multivariate analysis demonstrated that the effect of

Table 2. Summary of intra-operative parameters stratified by AKI.

	AKI	No AKI	p value
No. of grafts (mean $\pm$ SD)	3.0 $\pm$ 1.1	1.0 $\pm$ 1.1	.048*
Redo surgery <i>n</i> (%)	3 (33.3)	6 (66.7)	1.000
PRBC transfusion on-CPB (units) (mean $\pm$ SD)	4.67 $\pm$ 2.85	5.67 $\pm$ 7.5	.220
Use of side biter clamp <i>n</i> (%)	8 (72.7)	3 (27.3)	.023*
Duration of bypass (min) (mean $\pm$ SD)	200.2 $\pm$ 102.9	178.4 $\pm$ 87.3	.231
Duration of X-clamp (min) (mean $\pm$ SD)	144.7 $\pm$ 78.2	128.4 $\pm$ 71.1	.257
CPB MAP (average) (mmHg) (mean $\pm$ SD)	59.8 $\pm$ 6.8	59.9 $\pm$ 6.7	.938
CPB CI (average) (mmHg) (mean $\pm$ SD)	1.8 $\pm$ 0.2	1.8 $\pm$ 0.2	.336
CPB temperature (average) ($^{\circ}$ C) (mean $\pm$ SD)	34.5 $\pm$ 1.0	34.4 $\pm$ 1.0	.740
CUF (mL) (mean $\pm$ SD)	1424.5 $\pm$ 1124.7	984.6 $\pm$ 1045.9	.037*
MUF (mL) (mean $\pm$ SD)	164.8 $\pm$ 287.4	73.5 $\pm$ 160.8	.034*
Time < 60 MAP (min) (mean $\pm$ SD)	15.1 $\pm$ 9.5	15.5 $\pm$ 12.3	.885
Time < 50 MAP (min) (mean $\pm$ SD)	7.5 $\pm$ 3.7	9.4 $\pm$ 8.5	.390
Delta MAP (mmHg) (mean $\pm$ SD)	30.69 $\pm$ 9.82	28.17 $\pm$ 11.13	.223
Tranexamic acid (mg/kg) (mean $\pm$ SD)	36.8 $\pm$ 9.6	38.2 $\pm$ 9.9	.456

AKI: Acute Kidney Injury; PRBC: Packed Red Blood Cell; CPB: Cardiopulmonary Bypass; MAP: Mean Arterial Pressure; CI: Cardiac Index; CUF: Conventional Ultrafiltration; MUF: Modified Ultrafiltration.

**p* $\leq$.05.

Table 3. Serum NGAL values and AKI.

	AKI	No AKI	p value
	Median (IQR)	Median (IQR)	
NGAL baseline (ng/mL)	1437.7 (1357.4)	967.0 (1297.5)	.022*
6 h post-bypass (ng/mL)	5581.4 (5728.5)	1653.0 (2073.5)	.001*
Delta NGAL (ng/mL)	3795.00 (5218.5)	529.50 (937.45)	.001*
Percent rise in NGAL	74.21 (20.62)	33.83 (40.59)	.001*

NGAL: Neutrophil Gelatinase-associated Lipocalin; AKI: Acute Kidney Injury; IQR: Inter-quartile Range.

**p* $\leq$.05.

Table 4. Summary of intra-operative parameters stratified by AKI.

	AKI	No AKI	p value
Mortality <i>n</i> (%)	5 (11.4)	4 (5.9)	.312
IABP <i>n</i> (%)	0 (0.0)	3 (4.4)	.278
LCOS <i>n</i> (%)	13 (29.5)	16 (23.5)	.478
CRRT postop <i>n</i> (%)	1 (2.3)	1 (1.5)	1.000
CNS morbidity <i>n</i> (%)	8 (18.2)	4 (5.9)	.059
Respiratory system morbidity <i>n</i> (%)	8 (18.2)	6 (8.8)	.157
Wound infection <i>n</i> (%)	4 (9.1)	6 (8.8)	1.000
Re-exploration <i>n</i> (%)	2 (4.5)	1 (1.5)	.560
Arrhythmia <i>n</i> (%)	8 (18.2)	11 (16.2)	.782
Length of mechanical ventilation in days [median (IQR)]	2.0 (1.0)	2.0 (0.0)	.410
ICU stay in days [median (IQR)]	5.0 (3.0)	5.0 (1.0)	.554
VIS [(median $\pm$ IQR)]	10.0 (17.5)	10.0 (15)	.220

AKI: Acute Kidney Injury; IABP: Intra-Aortic Balloon Pump; LCOS: Low Cardiac Output Syndrome; CRRT: Continuous Renal Replacement Therapy; CNS: Central Nervous System; IQR: Inter-quartile range; ICU: Intensive Care Unit; VIS: Vasoactive Inotropic Score.

Table 5. Univariate analysis of significant variables.

Parameter	Odds ratio	95% CI	p value
Age (yrs)	1.042	1.011–1.074	.007*
BMI (kg/m ²)	1.112	1.012–1.223	.028*
Diabetes mellitus	3.654	1.324–10.088	.012*
Previous myocardial infarction	4.706	1.373–16.130	.014*
Clopidogrel	3.556	1.001–12.633	.050*
Valve surgery	1.558	0.872–2.785	.134
CABG valve	5.571	1.416–21.920	.014*
Number of grafts	1.898	1.043–3.454	.036*
CUF (mL)	1.000	1.000–1.001	.042*
MUF (mL)	1.002	1.000–1.004	.043*
Use of side - bite clamp	4.815	1.202–19.292	.026*

* $p \leq .05$.**Table 6.** Multivariate analysis and adjusted odds ratios.

Parameter	Odds ratio	95% CI	p value
Age (yrs)	1.024	0.989–1.060	.185
BMI (kg/m ²)	1.052	0.950 – 1.166	.330
Diabetes mellitus	1818058026.455	0.000	1.000
Previous myocardial infarction	1.789	0.048 – 66.575	.753
Clopidogrel	0.380	0.025 – 5.727	0.484
CABG valve	2232965765.668	0.000	1.000
Number of grafts	0.849	0.166 – 4.335	.844
CUF (mL)	1.000	1.000 – 1.001	.662
MUF (mL)	1.001	0.999 – 1.003	0.355
Use of side - bite clamp	0.000	0.000	1.000
Delta MAP (mmHg)	0.982	0.915 – 1.053	.606
Delta NGAL (ng/mL)	1.001	1.001 – 1.002	.002*

* $p \leq .05$.

perfusion pressure on AKI was attenuated, although it should be noted that they used a different model in their experiment, since their primary predictor of interest was oxygen delivery.⁷ Since on-bypass hemodilution and non-pulsatile flows blunt renal autoregulation, optimum blood pressure or flow to sustain renal oxygen delivery would be critical in the prevention of AKI.⁸ It is not clear which of these factors is most important, and whether a universal absolute lower threshold value (as we have used in this study) or an individualized relative change would be the major factor.

It is also noteworthy that, the limits of autoregulation are not the same in all individuals.^{9,10} Ngu et al., studied the impact of intraoperative hypotension on postoperative renal replacement therapy in patients undergoing cardiac surgery.¹¹ They stratified hypotension by MAP less than 55, 55 to 64, and 65 to 74 mmHg before, during, and after CPB. They found that each 10-min epoch of MAP less than 55 mmHg post-CPB was

associated with an adjusted odds ratio of 1.13 (95% CI, 1.05 to 1.23; $p = .002$), and each 10-min epoch of MAP between 55 and 64 mmHg post-CPB was associated with an adjusted odds ratio of 1.12 (95% CI, 1.06 to 1.18; $p = .0001$) for renal replacement therapy. In another study by Rettig et al., risk of AKI was found to be 1.07 to 1.11 times higher per 10 min of intraoperative hypotension on CPB in univariate analysis for all thresholds defined as a MAP of 50 mmHg or below and a drop in MAP of 60% from baseline ($p .01$).¹² We have analyzed the distribution of AKI (KDIGO criteria), over specified periods of low MAP on-bypass, by choosing two MAP thresholds, less than 60 mmHg and less than 50 mmHg. There was no statistically significant difference between the patients who developed AKI compared to those who did not demonstrate AKI when they were compared relative to the time spent on CPB at thresholds of either 50 mmHg (7.5 ± 3.7 min vs 9.4 ± 8.5 ; $p = .39$) or 60 mmHg (15.1 ± 9.5 min vs 15.5 ± 12.3 ; $p = .885$). de la Hoz

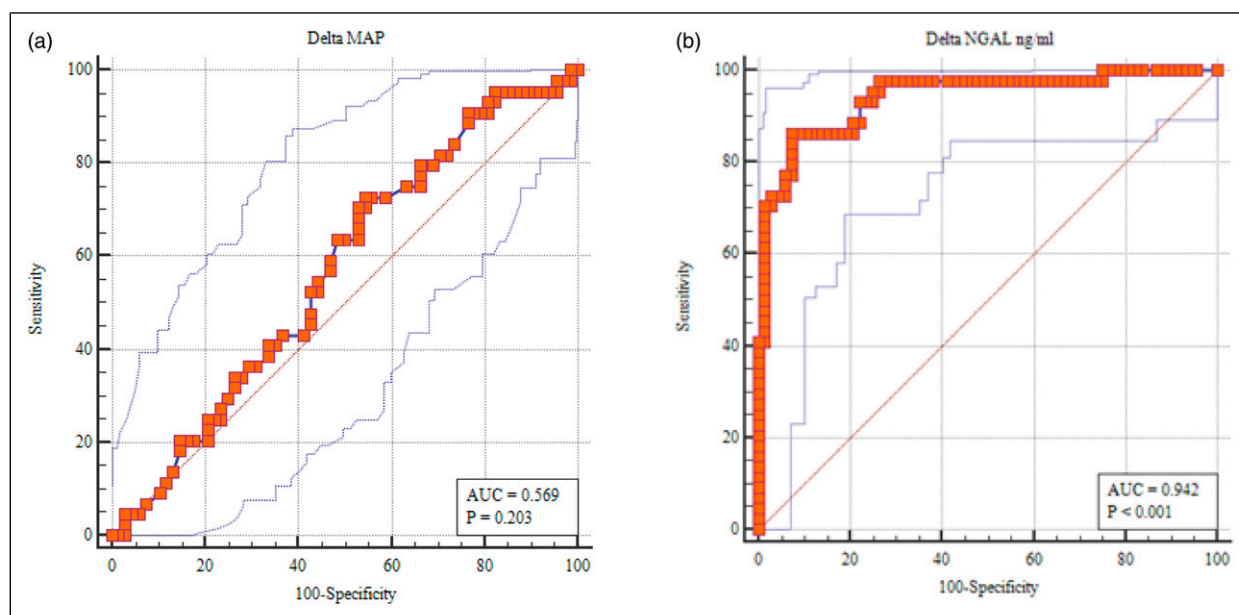


Figure 1. Receiver operating characteristic curves. (a) Delta MAP (b) delta NGAL. The cut-off value for delta NGAL was obtained at a sensitivity of 86.4% and specificity of 92.7%.

et al. found that their primary outcome of AKI was not associated with each 10-min period hypotension during cardiopulmonary bypass, fraction of hypotension during cardiopulmonary bypass of 60 to 80%, or total vasopressor-inotrope dose.¹³ With respect to the secondary outcomes, Al-Githmi et al. after a retrospective analysis stated that in their patients the most common risk factor associated with AKI following open-heart surgery was diabetes mellitus, followed by angina pectoris.¹⁴ Likewise, Sirvinskis et al., found that age; valve replacement, combined valve, and CABG surgery, duration of CPB, and aortic cross-clamp were the most important independent risk factors for ARF.⁶

In our study population, the two pre-operative characteristics which were found to be significantly associated with postoperative AKI were diabetes mellitus and previous myocardial infarction. As far as pre-operative medications were concerned, clopidogrel, had a significant correlation with postoperative AKI. The significance of clopidogrel in AKI is probably related to combined procedures, where many confounders related to CABG, might have played a role. We assessed various intra-operative factors and in combined valve and CABG surgeries, there was a correlation found between the number of grafts placed and AKI. Here, probably the number of grafts would imply a longer time on bypass, and more the number of diseased vessels handled, greater the chances of calcium emboli from atherosclerotic plaques getting mobilized into the circulation.

This might have an impact on blood flow to many vital organs including the kidney. Here, ischemia-reperfusion injury also plays a vital role. However, assessed independently, CPB time did not seem to affect the incidence of AKI in our patient population. Likewise, the placement of a side-biter clamp in combined surgeries was also found to be contributory. Whenever a side-biter clamp is placed on the aorta while doing proximal anastomosis, atherosclerotic plaques get disturbed and calcium embolizes into the circulation creating ischemic insult, thereby compounding the risk of AKI.²

Our study population was young, with an average age of 45.6 years in the AKI group and 38.2 years in the non-AKI group. Although there was a significant difference of univariate analysis between the groups, this was not seen on multivariate analysis. It is possible that the younger age group of our patients, as compared to previous literature done on the topic, could have influenced these findings. As we know, higher age is associated with loss of nephrons and diminution of baseline glomerular filtration rate (GFR), which could easily exacerbate the insults encountered on CPB, as compared to our younger population with healthier kidneys. Given the limitations of serum creatinine and its clearance, we measured the levels of serum NGAL and analyzed its predictive value for early CSA-AKI. In our study, ELISA test for human NGAL was performed on serum samples, and an ROC analysis estimated a cut-off value at more than 1661 ng/mL and found it to be

significantly associated with early CSA-AKI. Delta NGAL's sensitivity, specificity, and positive and negative predictive values were 86.4%, 92.7%, 88.4%, and 91.3%, respectively. Hence, our study highlights the reliability and predictive value of serum NGAL for early post-operative CSA-AKI.

Our study has the following limitations. Firstly, this is a single center study, with a fairly limited sample size, and a highly intuitive elucidation of AKI, spread over a fairly short post-surgical study period. This, on the whole, might have made our statistical power insufficient and prevented us from observing possible important variations in study parameters, such as extent of mechanical ventilation, extent of ICU stays and need for RRT. As previously mentioned, this was in part planned, in order to pinpoint, as best as realizable, the impact of surgical hemodynamic data on risk of post-surgical CSA-AKI. We sought to curb the effects of known confounders by employing an a priori range of principles and assembly of constituents that could help cause CSA-AKI. These aspects were involved in the multivariate analysis. Secondly, the study design is observational, making it prone to bias, and there was a specified plan a priori, which gives our analysis more of a post-hoc nature. Not being able to influence intra-operative decision-making may have resulted in situations arising, where cases who were deemed high-risk were taken better care of, during surgery, at a higher MAP. Thirdly, the surgeries in this cohort have been performed by different surgeons hence, cardiopulmonary bypass time varied, as did the complications, risks and surgical executions. Another factor was cardioplegia which was not uniform but changed as per the operating surgeon's choice. Our study patients received two types of cardioplegia, either of blood and del Nido. At last, we appreciate that for the aforementioned reasons, our single-center study of cases encountering valve/combined CABG + valve procedures with CPB at higher risk for CSA-AKI, has insufficient gross applicability, when taken to situations of cases at lower risk for CSA-AKI or those going through cardiac procedures at other hospitals or in other centers.

In conclusion, keeping in mind the limitations stated earlier, our study showed that delta MAP and CPB flow are not related to the phenomenon of early post-surgical CSA-AKI in cases with prior high-risk elements. We, however, found that serum NGAL, both at baseline, as well as the percent change during the early post-surgical period independently predicted the development of CSA-AKI. This implies that, there may be patients with a higher pre-operative preponderance to develop this complication, which could actually be delineated by the use of serum NGAL estimations at baseline. Further,

serum NGAL changes throughout the early peri-operative period could be used for tracking patients who might develop early CSA-AKI. Some of the other factors that we identified as being strongly associated with the occurrence of CSA-AKI are possibly discernable and modifiable. These include high body mass index, history of diabetes mellitus or uncontrolled sugars and previous myocardial infarction. We profess that these aspects necessitate supplementary consideration in future studies.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) disclosed receipt of the following financial support for the research, authorship, and/or publication of this article: This work was supported by the Jawaharlal Institute of Postgraduate Medical Education and Research; JIP/DEAN(R)/INTRAMURAL/PHS 1/2020-21 Dt 9/11/2020.

ORCID iD

Satyen Parida  <https://orcid.org/0000-0003-4752-3653>

References

1. Parida S, Badhe AS. Cardiac surgery-associated acute kidney injury. *J Anesth* 2013; 27(3): 433–446.
2. Kanji HD, Schulze CJ, Hervas-Malo M, et al. Difference between pre-operative and cardiopulmonary bypass mean arterial pressure is independently associated with early cardiac surgery-associated acute kidney injury. *J Cardiothorac Surg* 2010; 5: 71.
3. Ranucci M, Johnson I, Willcox T, et al. Goal-directed perfusion to reduce acute kidney injury: a randomized trial. *J Thorac Cardiovasc Surg* 2018; 156(5): 1918–1927.
4. Turner L, Hardikar A, Jose MD, et al. Acute kidney injury, stroke and death after cardiopulmonary bypass surgery: the role of perfusion flow and pressure. *Perfusion* 2021; 36(1): 78–86.
5. Slogoff S, Reul GJ, Keats AS, et al. Role of perfusion pressure and flow in major organ dysfunction after cardiopulmonary bypass. *Ann Thorac Surg* 1990; 50(6): 911–918.
6. Sirvinskis E, Andrejaitiene J, Raliene L, et al. Cardiopulmonary bypass management and acute renal failure: risk factors and prognosis. *Perfusion* 2008; 23(6): 323–327.
7. Newland RF, Baker RA, Woodman RJ, et al. Predictive capacity of oxygen delivery during cardiopulmonary bypass on acute kidney injury. *Ann Thorac Surg* 2019; 108(6): 1807–1814.
8. Baranauskas T, Kaunienė A, Švagždienė M, et al. The

- correlation of post-operative acute kidney injury and perioperative anaemia in patients undergoing cardiac surgery with cardiopulmonary bypass. *Acta Med Litu* 2019; 26(1): 79–86.
9. Hori D, Nomura Y, Ono M, et al. Optimal blood pressure during cardiopulmonary bypass defined by cerebral autoregulation monitoring. *J Thorac Cardiovasc Surg* 2017; 154(5): 1590–1598.e2.
 10. Ono M, Arnaoutakis GJ, Fine DM, et al. Blood pressure excursions below the cerebral autoregulation threshold during cardiac surgery are associated with acute kidney injury. *Crit Care Med* 2013; 41(2): 464–471.
 11. Ngu JMC, Jabagi H, Chung AM, et al. Defining an intraoperative hypotension threshold in association with de novo renal replacement therapy after cardiac surgery. *Anesthesiology* 2020; 132(6): 1447–1457.
 12. Rettig TCD, Peelen LM, Geuzebroek GSC, et al. Impact of intraoperative hypotension during cardiopulmonary bypass on acute kidney injury after coronary artery bypass grafting. *J Cardiothorac Vasc Anesth* 2017; 31(2): 522–528.
 13. de la Hoz MA, Rangasamy V, Bastos AB, et al. Intraoperative hypotension and acute kidney injury, stroke, and mortality during and outside cardiopulmonary bypass: a retrospective observational cohort study. *Anesthesiology* 2022; 136(6): 927–939.
 14. Al-Githmi IS, Abdulqader AA, Alotaibi A, et al. Acute kidney injury after open heart surgery. *Cureus* 2022; 14(6): e25899.